

Human Review: Uniformly convex ellipsoid with non-exposed extreme point

Original automatic proof: `solution_packet.pdf`

Checked date: 18 June 2026

Status: Manually checked.

Verdict: The proof appears correct.

Human Comments

The proof appears correct and clean. It gives a counterexample to the question in Remark 4.15 of Garcia-Lirola–Grelier–Rueda Zoca, and in fact proves a slightly stronger statement: the chosen point is not merely not strongly exposed by an ambient continuous functional; it is not exposed at all by such a functional.

The construction makes explicit the phenomenon already suggested in the source remark: the Minkowski functional of K may define a stronger norm than the ambient norm. The point is intrinsically strongly exposed for the Minkowski norm, but the corresponding norming functional is not continuous for the ambient Banach-space norm.

The Example

The ambient space is $X = \ell_2$, and

$$K = \left\{ x = (x_n) \in \ell_2 : \sum_{n=1}^{\infty} n^2 x_n^2 \leq 1 \right\}.$$

The Minkowski functional is the weighted Hilbert norm

$$|x|_K = \left(\sum_{n=1}^{\infty} n^2 x_n^2 \right)^{1/2}.$$

Thus K is the unit ball of a Hilbert norm on $\text{span } K$. Since

$$\|x\|_2 \leq |x|_K,$$

uniform convexity of the Hilbert ball implies uniform convexity of K as a subset of the ambient space ℓ_2 .

The chosen point is

$$x_0 = \left(\frac{c}{n^2} \right)_{n \geq 1}, \quad c = \left(\sum_{n=1}^{\infty} \frac{1}{n^2} \right)^{-1/2}.$$

Then $|x_0|_K = 1$, so x_0 is an extreme point of K , because K is the unit ball of a strictly convex Hilbert norm.

Why It Is Not Exposed

Let an ambient continuous functional be represented by $b = (b_n) \in \ell_2$:

$$\varphi_b(x) = \sum_{n=1}^{\infty} b_n x_n.$$

With respect to the weighted Hilbert inner product

$$\langle x, z \rangle_K = \sum_{n=1}^{\infty} n^2 x_n z_n,$$

this functional is represented by

$$z = \left(\frac{b_n}{n^2} \right)_{n \geq 1}.$$

If φ_b exposed x_0 over K , then equality in the Hilbert-space Cauchy-Schwarz inequality would force

$$z = \lambda x_0$$

for some $\lambda > 0$. Since $x_{0,n} = c/n^2$, this implies

$$\frac{b_n}{n^2} = \lambda \frac{c}{n^2}, \quad \text{hence} \quad b_n = \lambda c$$

for every n . But a nonzero constant sequence is not in ℓ_2 . Thus no ambient continuous functional can expose x_0 .

Therefore x_0 is not exposed, and hence is not strongly exposed.

Review Outcome

Approve this packet as an apparently correct counterexample. The proof is short, transparent, and directly illustrates the distinction between exposure for the intrinsic Minkowski functional and exposure by a functional continuous for the ambient norm.